

Asadullah Farooqi (Asadullah)

Senior Software Engineer | Backend & Infrastructure Specialist

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years building and scaling distributed systems, payment infrastructure, and cloud-native applications. Proven track record leading engineering teams, implementing high-availability backend services, and establishing DevOps/SRE practices. Expertise in Golang microservices, Kubernetes orchestration, AWS infrastructure, and data pipeline engineering. Successfully automated 99% of reconciliation processes and improved system observability by 100% through Elasticsearch integration.

Core Competencies: Distributed Systems • Microservices Architecture • Site Reliability Engineering • Cloud Infrastructure (AWS) • Container Orchestration • Data Engineering • Technical Leadership

TECHNICAL SKILLS

Languages: Golang (Expert) • Python (Advanced) • Rust (Intermediate) • Bash/Shell Scripting

Infrastructure & DevOps: Kubernetes • Docker • AWS (EKS, RDS, SQS, SNS, Aurora) • Azure IoT • Terraform • GitHub Actions

Backend & Data: MySQL • PostgreSQL • MongoDB • Redis • Elasticsearch • ETL Pipelines • Event-Driven Architecture

Protocols & APIs: gRPC • REST • GraphQL • MQTT • Protocol Buffers

Frameworks: Go-kit • Gin • Echo • Django • Flask • Tokio

SRE & Observability: Monitoring • Logging • Alerting • Distributed Tracing • Performance Optimization

PROFESSIONAL EXPERIENCE

Senior Software Engineer | [SafePay](#) | Apr 2022 - Present

Leading backend infrastructure and platform engineering for fintech payment systems

Technical Leadership & Team Management

- Lead a team of engineers in designing and delivering scalable payment infrastructure serving high-volume transaction processing
- Mentor junior engineers on system design, code quality, and production reliability best practices
- Drive technical decision-making for microservices architecture and infrastructure choices

Infrastructure & Platform Engineering

- **Observability & Analytics:** Architected and deployed Elasticsearch across the entire microservices ecosystem with real-time event listeners and historical backfillers, improving operational visibility and analytics capabilities by 100%
- **Automation & Reliability:** Designed and implemented a controllable async reconciliation system in Golang, achieving 99% automation of financial reconciliation processes and reducing manual intervention from hours to minutes
- **Data Infrastructure:** Built production data warehouse on PostgreSQL from the ground up, including ETL data pipelines, Kubernetes batch jobs for backfilling, and SQS-based event streaming architecture
- **Audit & Compliance:** Implemented a comprehensive audit logging system for admin operations, ensuring compliance and traceability

Backend Systems Development

- Developed an advanced settlement generation system supporting marketplace business models (Stripe Connect-style architecture)

- Built microservices from scratch for payment gateway integrations (Easypaisa, AbhiPay, card payments) using Golang, Go-kit, gRPC, with optimized MySQL database schemas
- Created a utility billing service covering 90% of national service providers with high availability requirements
- Engineered reporting APIs and complex SQL queries (MySQL/PostgreSQL) for real-time merchant and admin dashboards

Security & API Development

- Extended core services with additional security layers, validation middleware, and BIN-based MDR charging mechanisms
- Developed gRPC clients and Protocol Buffer definitions for inter-service communication
- Implemented RESTful APIs for payment processing with comprehensive error handling and retry logic

Tech Stack: Golang, Kubernetes, Docker, AWS (EKS, RDS, SQS, SNS), Go-kit, MySQL, PostgreSQL, ElasticSearch, gRPC, Event-Driven Architecture

Software Engineer | [Siemens DISW](#) | Feb 2020 - Mar 2022

Embedded systems and IoT infrastructure development

IoT Platform Development

- Integrated AWS IoT Core and Azure IoT Hub into an edge IoT solution, implementing MQTT-based telemetry, device provisioning, authentication, and OTA firmware updates using cloud SDKs
- Developed features for a production-embedded IoT framework in C, focusing on resource-constrained environments

DevOps & Testing Infrastructure

- Built a comprehensive test automation framework in Python from scratch, covering unit tests, integration tests, and firmware update validation
- Reduced manual testing time and improved release confidence through automated testing pipelines

Tech Stack: C, Golang, Python, AWS IoT, Azure IoT, Docker, Linux, GCC/Delve Debuggers, Make, CMake, MQTT

Contract Software Engineer | Various Clients | Dec 2019 - Feb 2023

[BP8 Fitness](#) - Backend engineer implementing features in Golang with React frontend, MongoDB, and Docker

[Halidai](#) - Built a social media web app with Django, CDN integration for media storage, and MongoDB

[Punch Group](#) - Developed checkout frontend for [Skipify](#) using Next.js

[Punch Group](#) - Created a Flask web app automating sales workflow and optimized scrapers, reducing cloud costs by 40%

Tech Stack: Golang, Python, Django, Flask, React, Next.js, MongoDB, Docker

Software Engineering Intern | Maher Inc. ([StyleMePretty](#)) | Dec 2018 - Mar 2019

- Resolved frontend bugs and UI inconsistencies across the platform
- Built an analytics dashboard with backend statistics APIs (Django) and interactive charts (EChart.js) for business metrics visualization
- Improved data presentation and decision-making capabilities for stakeholders

Tech Stack: Python, Django, AngularJS, React, Pandas, Docker

PROJECTS & ACHIEVEMENTS

[Inventory Management Web App](#) (Nov 2017 - Mar 2018)

First production project: Built a responsive inventory management system for a multi-branch business with ledger tracking. Deployed on Pythonanywhere with Gunicorn WSGI server.

EDUCATION

FSc (Mathematics) | Government College Loralai | 2013-2015

Matriculation (Science) | Farooqi Public High School | 2011-2013

LANGUAGES

English (Fluent) • Urdu (Fluent) • Pashto (Fluent) • Arabic (Proficient) • Persian (Beginner)

ADDITIONAL INFORMATION

- **Interests:** Fleet management systems, Linux internals, low-level operating systems, distributed systems architecture, Rust, Elixir
- **Open to:** Remote, Hybrid, or On-site positions | Full-time, Permanent roles
- **Availability:** Immediate / 1 Month notice period